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Uncertainty, FDI Inflows, and Financial Market Development: Empirical Evidence

Godfrey Marozva * and Margaret Rutendo Magwedere

Department of Finance, Risk Management and Banking, University of South Africa, Pretoria P.O. Box 392, South Africa; magwemr@unisa.ac.za
* Correspondence: godezhy@gmail.com

Abstract: This study offers insight into the multifaceted relationship between economic policy uncertainty, foreign direct investment, and financial development. It analyzes the effects of economic policy uncertainty and financial development on foreign direct investment (FDI) inflows, using a sample of 75 emerging markets and developing countries, over 2000–2023. To achieve this, a pooled mean group (PMG) was used, and the finding of this study for this period is a positive long-term relationship between economic policy uncertainty and foreign direct investments inflows, challenging the traditional view that uncertainty deters investment. In the long term, the joint effects of financial development and economic policy reduced foreign direct investments. These findings highlight the critical role of financial development in shaping the long-term impact of uncertainty on FDI, and they underscore the need for policy to enhance financial systems in order to stabilize and sustain FDI inflows in uncertain environments.

Keywords: foreign direct investment; economic policy uncertainty; financial development; stability; panel data; auto regressive distributed; cointegration

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1. Introduction

Stable and predictable policies are essential for foreign direct investment, and more so in emerging and developing economies. In emerging and developing economies, foreign direct investments (FDI) are recognized as a catalyst for economic growth and development (Aust et al., 2020). Foreign direct investment is essential for emerging markets as it is an engine of growth, providing access to capital, technology, and expertise that can boost productivity and economic development (Emako et al., 2023). Attractive extractive industries, population growth, and rising consumption patterns have remained a magnet for foreign investors in emerging and developing economies (Adegboye & Okorie, 2023). However, capital flight to safer destinations with stable economic policies remains an alternative trade-off for investors (Nguyen & Lee, 2021). Under periods of uncertainty, investment capital tends to fly to quality, with effects on the financial development of domestic economies. Economic policy uncertainty increases the volatility of capital flows such that in periods of high economic policy uncertainty, countries with weak institutional frameworks can experience lower capital inflow. Studies on uncertainty have mainly been more available on the micro level (i.e., the firm level) rather than on the macro level, mainly due to the non-availability of data on uncertainty measures at the macro level and across countries.

Most emerging and developing markets' EMDEs have very low entry points in the debts and equity markets, which offer attractive entry points for investors (Amoroso & Müller, 2018). The market capitalization in most EMDEs is lower, offering potential

for growth to most investors. Despite this opportunity for investors in emerging and developing markets, the potential for foreign direct investment inflows remains very low in these economies. Notwithstanding EMDEs being endowed with natural resources and cheap labor, the marginal productivity of capital remains very low and fails to attract higher FDI given the potential for higher investment yields (United Nations, 2024).

However, FDI inflows are influenced by several factors, which include the stability of the policy environment and the specific country's level of financial development. Economic policy uncertainty (EPU), which refers to the unpredictability of government actions related to fiscal, monetary, and regulatory policies, can deter foreign investors by increasing perceived risks and lowering the attractiveness of investment opportunities. On the other hand, financial development, characterized by a well-functioning financial system (including banking sectors, capital markets, and access to credit), can facilitate FDI by reducing transaction costs, improving access to financing, and providing a more efficient allocation of resources (Desbordes & Wei, 2017). Financial intermediation costs in some emerging and developing markets are higher, and they contribute to the foreign direct investment paradox. Despite the labor and natural resources endowments, the continent is not able to attract financial capital inflows. The financial sectors in most of these countries are underdeveloped. The United Nations (2024) suggested that inward foreign direct investment in EMDEs is a paradox as capital is expected to flow from countries with low returns to countries with high returns. However, in Africa, despite the potential for higher returns on investment, FDI inflows remains low compared with other regions of the world (Adegboye & Okorie, 2023). A well-developed financial system may help cushion the negative effects of policy uncertainty by providing foreign investors with easier access to financing, hedging opportunities, and a more liquid market. Conversely, in countries with low levels of financial development, the impact of EPU may be more pronounced as investors face additional barriers to investing in uncertain environments.

While a growing body of literature has explored the bilateral relationships between economic policy uncertainty (EPU), foreign direct investment (FDI), and financial market development (FD), most studies tend to examine these linkages in isolation, neglecting the complex, interdependent dynamics among them. Specifically, the role of uncertainty as a systemic determinant of both capital flows and financial sector performance remains underexplored, particularly in the context of its moderating effect on the FDI–financial development nexus. This has mainly been due to the lack of data on the EPU measure, which is continuously improving. Moreover, existing empirical research often overlooks the interactive effect of financial development and uncertainty (FD $\times$ EPU) in influencing economic outcomes, and few studies adopt panel error correction frameworks (such as PMG models) to capture the long- and short-term asymmetries across diverse country settings. This study addresses these gaps by investigating the triangular relationship between EPU, FDI inflows, and financial development, while introducing a novel interaction term (FD_EPU) to empirically assess how uncertainty conditions influence the effectiveness of financial markets in absorbing and transmitting FDI. The originality of this study lies in its integrated empirical design, the use of dynamic panel techniques, and its emphasis on the conditional role of uncertainty, offering new insights for both theory development and policy design in emerging and developed economies alike.

Despite the widely acknowledged view that EPU negatively affects FDI, it is unclear how the level of financial development influences this relationship; that is, the interaction between EPU and financial development is less well understood. This study seeks to provide empirical analysis on the long-term impact of financial sector development and uncertainty on foreign direct investment inflows. Moreover, this study seeks to answer the questions of how economic policy uncertainty (EPU) affects the foreign direct investment of

host countries, as well as that of the role of local financial markets in mediating the potential benefits. A. Dixit (2011) argued that foreign direct investment is the form of capital flow that is most exposed to the host countries' political and institutional uncertainty. Although emerging markets and developing economies are attractive to foreign investors because of their higher investment yields, the flow of capital into these states remains low. Previous studies have mainly focused on the long-term determinants of FDI; however, events such as the global financial crisis, Brexit, the COVID-19 pandemic, and the Russian annexation of Ukraine, among others, shows that short-term effects such as uncertainty can have a significant effect on FDI inflows. This study also seeks to expand the literature on the determinants of FDI in emerging markets and developing economies.

This study is organized as follows: the theoretical and empirical literature underpinning the article are discussed in Section 2; our methodology is presented in Section 3, in which we describe the data and the model specification; the results of the net foreign direct investment inflows and economic policy uncertainty are presented in Section 4, in which we further analyze the moderating role of financial development; and Section 5 presents a discussion of the results and concludes this study, providing a summary of the policy implications.

2. Literature Review

In this section, a detailed exploration into the theoretical and empirical literature on foreign direct investment, economic policy uncertainty, and financial development is provided. This section also discusses the mediating role of financial development with respect to FDI. There is an expectation that economic policy uncertainty negatively affects foreign direct investment.

2.1. Theoretical Literature

In examining the long- and short-term relationships between foreign direct investment inflows, economic policy uncertainty, and financial development, this study is guided by the real options theory (ROT). The real options theory estimates an investment project that is dependent on real assets to support a firm's decision process or capital budget planning (A. K. Dixit & Pindyck, 1994; Trigeorgis, 1996). This theory assumes that there are three main factors to consider in investment decision-making; these are the irreversibility of investments, uncertainty, and the timing of investments (Carruth et al., 2000). Real options theory brings the theory of financial options from the capital markets to the realm of investment and strategic decision-making under conditions of uncertainty (Trigeorgis, 1996). The role of uncertainty in affecting foreign direct investment has also been affirmed by Bernanke (1983). High uncertainty encourages firms to delay investment when the investments are irreversible or costly to reverse. Theorists have equated the investment decision to exercising a financial option where high uncertainty increases the value of waiting (Stokey, 2016). The ROT states that investors have the right, but not the obligation, to take a specific future investment decision at a specified cost (Trigeorgis, 1996; Bloom et al., 2007). Thus, investors can delay, invest, or abandon investments based on changing economic and institutional conditions (Bernanke, 1983; Baker et al., 2016). Delaying investments allows one to avoid committing capital in an environment where future returns are unclear. Under these conditions, the inflow of foreign direct investment is affected. Thus, favorable and stable economic policy environments reduce the waiting period for investors and encourage increased inflows of foreign direct investments (Bernanke, 1983). The previous theoretical framework on FDI based on ownership, location, and internationalization (OLI) by Dunning (1977, 1979) was criticized because location and internationalization are thought to fall away in periods of high economic policy uncertainty (Hsieh et al., 2019).

The key observation in the theory is that periods of high and sustained economic policy uncertainty may result in delays to, or the abandonment of, foreign direct investments (Zhang et al., 2023). There is no unified theory on foreign direct investment, economic policy uncertainty, and financial development.

2.2. Empirical Literature

Although foreign direct investment is tipped to be the most stable of the foreign capital flows, it is also highly susceptible to economic policy uncertainty and high fixed costs (Choi et al., 2021). Bloom (2009) argued that uncertainty increases the waiting costs for investors. High EPU leads investors to delay or reduce FDI as they prefer to wait for clearer policy signals before capital commitments; high policy uncertainty means higher perceived risk, and investors postpone entering markets with higher perceived risks (Nguyen & Lee, 2021). This increases the cost of capital and reduces the attractiveness of long-term investments such as FDI (Choi et al., 2021). Despite the economic benefits of foreign direct investment, such as economic growth and technological transfer, among others, uncertainty increases the complexity of FDI in emerging and developing economies (Hsieh et al., 2019). Uncertainty increases the complexity of doing business for multinational enterprises (MNEs) due to challenges in predicting the future trends in government policy and economic outcomes (Bloom et al., 2007).

In other studies, financial development is argued as the channel through which policy uncertainty affects foreign direct investment inflows (Nguyen & Lee, 2021). Hence, this study also examined the mediating effects of the financial development of the FDI–EPU nexus. Choi et al. (2021) argued that financial development is the most important channel that affects foreign direct investment inflows. In this regard, it is thought that countries with developed financial markets offer investors' alternative access to funding and risk management options (Canh et al., 2020); a well-developed financial system mitigates the effects of economic policy uncertainty (Karaman & Yıldırım-Karaman, 2019; Choi et al., 2021). This implies that higher financial development has the potential to increase FDI as investors have increased access to capital markets; thus, they have access to insurance to hedge against risks, reducing the option value of delaying or abandoning an investment decision (Li & Rugman, 2007; Canh et al., 2020). However, it is expected that a period of higher uncertainty has the potential to lower foreign direct investment inflows. Uncertainty is not only associated with downside risks but also with potential future opportunities (Li & Rugman, 2007). In most previous studies, the focus on the determinants of foreign direct investment was mainly centered on transaction cost economics, with few considering the role of uncertainty, mainly due to lack of data on uncertainty measurement (Azzimonti, 2019; Nguyen et al., 2018; Honig, 2020; Choi et al., 2021).

There is an acknowledgement that among the determinants of FDI, short-term effects such as uncertainty affect FDI inflows (Jardet et al., 2023). Previously, a lack of cross-country uncertainty measures has limited the empirical studies on uncertainty as a determinant of FDI for a number of countries. These challenges regarding the cross-country measure of uncertainty motivated this study, as there are currently data improvements in the measure of uncertainty (Bloom, 2009) A. Dixit (2011) argued that FDI are the most abundant capital flows exposed to economic, political, and institutional uncertainty. Thus, global risk aversion by investors can result in flight to quality, affecting the level of FDI in a country for a specific period. Recent previous studies appear to assume linearity in the relationship between the determinants of FDI (Nguyen & Lee, 2021; Choi et al., 2021; Avom et al., 2020), save for An and Yeh (2021) who argued for the possibility of a nonlinear relationship, and more so when financial development is considered as a determinant of FDI.

Given that the aim of this study is to examine the long- and the short-term effects of economic policy uncertainty on foreign direct investment (net inflows), we present the following hypotheses:

H1. *EPU has no significant impact of the net inflows of foreign direct investment.*

H2. *The mediating effects between financial development and economic policy uncertainty (FD × EPU) have no significant impact on foreign direct investment inflows.*

3. Methodology

Data and Variables

This section describes the data and outlines the empirical strategy regarding the long- and short-term effect and the causal effects of economic policy uncertainty and financial development on foreign direct investment. This study used panel data from 2000 to 2023 for 75 emerging and developing markets. The selection of the study period and the countries included was mainly determined by data availability.

Due to the non-availability of data on the financial markets in some of the emerging and developing markets, this study uses only banking data, rather than also including the data from the stock market. Table 1. Presents the variables, definition of variables and the sources of data.

Table 1. Variables, definition of variables, and data sources.

Variables	Definition of Variables	Source	Expected Sign
Foreign direct investment inflows (FDI)	Foreign direct investment; net inflows (% of GDP)	World Development Indicators (WDI)	
Economic policy uncertainty (EPU)	Risk associated with undefined future government policies and regulatory frameworks	https://www.policyuncertainty.com (accessed on 19 May 2025)	-
Financial development (FD)	Domestic "credit to the private sector by banks (% of GDP)"	WDI	+/−
EPU*FinDev	The interaction between EPU and FinDev	WDI and https://www.policyuncertainty.com (accessed on 19 May 2025)	+/−

Source: authors' compilation.

In this study, the foreign direct investment is hypothesized to be a function of uncertainty, financial development, and the interaction term between financial development and uncertainty:

$$FDI = f(EPU, FinDev, EPU{*}FinDev) \quad (1)$$

where FDI is foreign direct investment inflows; EPU is the economic policy uncertainty index, FinDev is the financial development; and (EPU∗FinDev) is the interaction term between economic policy uncertainty and financial development.

This study employed the panel autoregressive distributed lags (ARDL) methodology as the preferred estimation technique. The choice of this methodology was mainly centered on its simplicity and its ability to capture both short- and long-term periods simultaneously in analysis. Following Pesaran et al. (1999), the panel ARDL is the preferred method for examining the cointegrating the relationships between the variables as variables of different orders and degrees of integration can be used in the estimation as long as they do not have an integration of more than 2 that is I(2). As a prerequisite to any economic analysis,

stationarity tests of each of the variables were employed to determine the variables' order of integration.

As argued by Pesaran et al. (2001), Narayan (2004), and Magwedere and Marozva (2024), panel ARDL is appropriate in small samples, and it also concurrently examines the long- and the short-terms relationships between the variables. Before the panel model is estimated, this study checked the stationarity of the variables using the unit root tests and the results are presented in Appendix A. The empirical literature recommends the Levin et al. (2002), Breitung and Das (2005), Im et al. (2003), Maddala and Wu (1999), and Hadri (2000) panel unit root tests. The mean group (MG), dynamic fixed-effects (DFE) group, and pooled mean group (PMG) estimations methods were compared to determine the best-fitting model for the data that accounts for the efficiency and consistency of the estimators. The Hausman test was used to select a robust model for the data. Given the variables under consideration in this study, the panel ARDL is specified as follows:

$$\begin{gathered}\Delta FDI_{it} = \varnothing_i(FDI_{i,\,t-1} - \gamma_{1i}EPU_{i,t} - \gamma_{2i}FinDev_{i,t} - \gamma_{3i}EPU{*}FinDev_{i,t}) + \\ \sum_{j=0}^{q-1}\beta_{1i}\Delta EPU_{i,t-j} + \sum_{j=1}^{p-1}\delta_{ij}\Delta FDI_{i,\,t-j} + \sum_{j=0}^{q-1}\beta_{2i}\Delta FinDev_{i,t-j} + \\ \sum_{j=0}^{q-1}\beta_{3i}\Delta EPU{*}FinDev_{i,t-j} + \mu_i + \varepsilon_{it}\end{gathered} \tag{2}$$

$$\begin{gathered}\Delta EPU_{it} = \varnothing_i(EPU_{i,\,t-1} - \gamma_{1i}FDI_{i,t} - \gamma_{2i}FinDev_{i,t} - \gamma_{3i}EPU{*}FinDev_{i,t}) + \\ \sum_{j=0}^{q-1}\beta_{1i}\Delta EPU_{i,t-j} + \sum_{j=1}^{p-1}\delta_{ij}\Delta FDI_{i,\,t-j} + \sum_{j=0}^{q-1}\beta_{2i}\Delta FinDev_{i,t-j} + \\ \sum_{j=0}^{q-1}\beta_{3i}\Delta EPU{*}FinDev_{i,t-j} + \mu_i + \varepsilon_{it}\end{gathered} \tag{3}$$

$$\begin{gathered}\Delta FinDev_{it} = \varnothing_i(FinDev_{i,\,t-1} - \gamma_{1i}FDI_{i,t} - \gamma_{2i}EPU_{i,t} - \gamma_{3i}EPU{*}FinDev_{i,t}) + \\ \sum_{j=0}^{q-1}\beta_{1i}\Delta FinDEv_{i,t-j} + \sum_{j=1}^{p-1}\delta_{ij}\Delta FDI_{i,\,t-j} + \sum_{j=0}^{q-1}\beta_{2i}\Delta EPU_{i,t-j} + \\ \sum_{j=0}^{q-1}\beta_{3i}\Delta EPU{*}FinDev_{i,t-j} + \mu_i + \varepsilon_{it}\end{gathered} \tag{4}$$

FDI_{it} represents foreign direct investment net inflows for country i for year t; EPU_{it} is the economic policy uncertainty for country i for year t; $FinDev_{it}$ is the financial development for country i at time t; $EPU{*}FinDev_{it}$ is the interaction between economic policy uncertainty for country i at time period t; whilst μ_i and $\varepsilon_{i,t}$ denote the fixed effects and the error term, respectively. $\beta_{1\ldots N}$, δ, are the short-term coefficients; the long-term coefficients in the model are represented by γ_{1i} and γ_{2i}; whilst $\varnothing_i$ represents the speed of adjustment to the long-term equilibrium. This study controlled for fixed effects to mitigate the concern that policy uncertainty is often correlated across countries or driven by a common factor.

Given the models specified in Equations (2)–(4), the conceptual framework of the study is outlined in Figure 1.

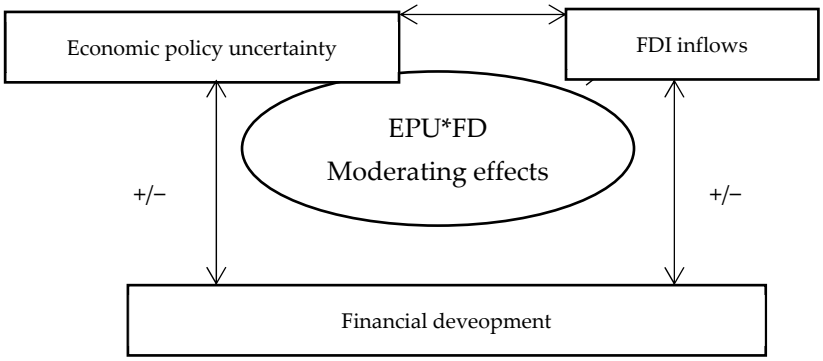


Figure 1. Conceptual framework. Source: authors' own design.

The conceptual framework depicted in Figure 1 illustrates the interplay between economic policy uncertainty, financial development, and foreign direct investment (FDI) inflows. It theorizes that increases in economic policy uncertainty can deter FDI inflows, while robust financial development may attract such investments. Furthermore, the framework suggests that financial development serves as a mediating channel through which economic policy uncertainty impacts FDI inflows.

4. Results

The descriptive statistics, correlation analysis, unit root tests, and the results from the panel ARDL estimation are presented in this study.

4.1. Descriptive Statistics

The descriptive statistics of this study are presented in Table 2.

Table 2. Descriptive statistics.

Variable	Obs	Mean	Std. Dev	Min	Max
FDI	1800	3.2883	4.0354	−37.1727	43.9121
EPU	1800	0.0646	0.0512	0	0.4180
FD	1798	32.4364	28.4516	0.0015	263.5903
FD_EPU	1799	2.0871	2.6866	0	25.1127

Source: authors' own computations.

Table 2 shows the descriptive statistics of the study. The maximum for FDI as a percentage of the respective countries' GDP is 43.91%, indicating notable FDI inflows across the countries with an average of 3.288. The foreign direct investment for the countries is closely clustered around the average of the dataset, as the standard deviation for FDI is 4.0354. For the EPU, the maximum is 0.4180, with a minimum of 0 and a standard deviation of 0.0512. The average of the EPU is 0.0646. Financial development has an average of 32.4364, with a much larger standard deviation of 28.4516, suggesting a high degree of variability in financial development across the countries. The values range from a minimum of 0.0015 to a maximum of 263.5903, suggesting that financial development levels differ widely across the observations. The interaction between financial development and economic policy uncertainty (FD_EPU) has a mean of 2.0871 and a standard deviation of 2.6866, displaying a rational degree of variation. The values range from 0 to 25.1127, indicating that the influence of EPU on financial development varies significantly.

4.2. Empirical Results

The empirical results on the nexus betweed FDI, EPU, FD and the interaction between FD and EPU are presented in Table 3.

Table 3. Panel ARDL and ECT results.

	PMG	PMG	PMG	PMG
Variables	**D.FDI**	**D.EPU**	**D.FD_EPU**	**D.FD**
Long-term				
L.EPU	3.970 *		0.0371 ***	12.98 **
	(1.736)		(0.000479)	(4.435)
FD_EPU	−0.0911 *	−0.000900		329.5 ***
	(0.0411)	(0.00248)		(80.72)

Table 3. *Cont.*

	PMG	PMG	PMG	PMG
Variables	**D.FDI**	**D.EPU**	**D.FD_EPU**	**D.FD**
FD	0.00111 (0.00341)	0.000206 (0.000170)	−0.00172 *** (0.0000479)	
FDI		−0.000482 (0.000389)	0.000636 *** (0.000128)	−0.102 (0.0623)
ECT	**−0.533** *** (0.0339)	**−0.560** *** (0.0253)	**−0.239** *** (0.860)	**0.227** *** (0.0260)
Short-term				
D.EPU	−1.497 (1.472)		−4.512 *** (1.270)	8.938 (9.270)
D.FD_EPU	62.06 (62.02)	−0.471 (0.475)		74.57 *** (8.445)
D.FD	−15.68 (15.68)	−0.0834 (0.0837)	0.0570 *** (0.00587)	
D.FDI		−0.000718 (0.000886)	0.0241 (0.0266)	−0.251 (0.294)
_cons	1.539 *** (0.142)	−0.0323 *** (0.00211)	0.830 *** (0.0654)	−6.013 *** (1.255)
N	1647	1647	1647	1647

The robust standard errors are in parenthesis. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Source: authors' own computations using Stata 15.1.

5. Discussion

The results of the long-term and the short-term relationship between economic policy uncertainty, foreign direct investment, and financial development are presented in Table 3. The results also present findings on the effect of the interaction between economic policy uncertainty and financial development on foreign direct investment.

This study found a positive long-term relationship between economic policy uncertainty and foreign direct investment. The null hypothesis of no significant effect of economic policy uncertainty to net inflows in FDI is rejected at a 5% significance level. The a priori expectations were a negative relationship between the variables, as higher economic policy uncertainty is expected to deter foreign direct investment. This can possibly reflect a "risk-taking" or speculative investment behavior in uncertain environments. The real options theory argues for the irreversibility of an investment decision, and possibly, in most cases, the firms have already invested. Thus, there is a possibility of increased investments irrespective of higher uncertainty. Risk-neutral investors may increase their investments during periods of higher uncertainty (Miao et al., 2021). Economic policy uncertainty positively affects FDI and FD_EPU in the long term but can be destabilizing in the short term. The theory of strategic investment (first-mover advantage) can explain the positive relationship between foreign direct investment and economic policy uncertainty. In uncertain economic environments, some firms might make strategic investments to establish their presence in a new market before their competitors (Struckell et al., 2022). This "first-mover advantage" can give foreign firms a foothold in the market when local firms are hesitant or constrained by uncertainty. By entering the market during uncertain times, foreign investors may secure a dominant market position once the uncertainty subsides. The positive relationship between uncertainty and FDI in these settings could stem from the perception that uncertain markets provide high potential return and opportunities, especially in industries where early entry or monopolistic advantages can be gained. In

such cases, foreign firms may be drawn by prospects of greater returns despite, or even because of, the higher risks, and more so in economies with higher economic growth (Okafor et al., 2022).

There are mixed findings in the literature concerning the relationship between EPU and foreign direct investment. Choi et al. (2021)'s findings suggested that the effect of EPU on FDI is stronger in countries with less financial development. However, for this study, financial development moderates the relationship between EPU and FDI. It is possible that financial development offers risk management instruments, such as hedging tools and diversified financial products, which can help foreign investors manage the risks associated with high EPU (Zhang et al., 2023). The financial development and economic policy uncertainty interaction (FD_EPU) negatively affects FDI (−0.0911 *) in the long term. This could imply that the unchecked amplification of financial volatility due to policy shocks may discourage stable foreign capital inflows.

The joint effects of financial development and economic policy uncertainty is negative and significant. This rejects the null hypothesis of no relationship between the FDI and the mediating effects of FD and EPU. Furthermore, the findings of a negative and significant relationship between FDI and the joint effects of financial development and policy uncertainty concur with the findings of Nguyen and Lee (2021). Economic policy uncertainty and foreign direct investments have strong positive long-term effects of 0.0371 and 0.000636 on the interaction between economic policy uncertainty and financial market development. In a separate argument, Inada and Jinji (2024) suggested that the role of policy uncertainty in foreign direct investments depends on international investment agreement (IIA), which mitigates policy uncertainty by increasing FDI inflows. Additionally, firms that possess a competitive edge in navigating uncertain markets, such as those with superior risk management capabilities or experience in volatile regions, may view uncertain conditions as entry points through which to secure favorable terms or dominate less competitive markets.

For Equations (1)–(3), the error correction terms are all negative and significant (−0.533 ***, −0.560 ***, −0.239 ***), showing that the system adjusts to its long-term equilibrium.

6. Conclusions

This study sought to provide an understanding of uncertainty (risk), foreign direct investment, and financial development in emerging and developing markets. Contrary to the traditional view that uncertainty deters foreign investment, this study suggests that in some cases, economic policy uncertainty may increase FDI inflows. This positive relationship can occur when foreign investors perceive potential opportunities in volatile environments, particularly in sectors where first-mover advantages or high returns are achievable, despite the risks. However, in the long term, the interaction effects show that low financial development amplifies the volatility of this relationship, and the long-term relationship is negative. It is possible that low financial development limits the risk management options for investors. Furthermore, the interaction between financial development and uncertainty (FD_EPU) emerges as a crucial transmission channel, highlighting the need for adaptive, integrated, and risk-aware policy frameworks. These findings emphasize that economic policy uncertainty must be actively managed through strategic financial development guidance to ensure that it contributes to both attracting sustainable FDI and enhancing macroeconomic stability.

Policymakers should therefore adopt a holistic approach that coordinates financial, investment, and institutional reforms to reinforce long-term economic growth in an increasingly uncertain global environment. The main limitation of this study is that only banking sector data were available for a number of cross sections. Stock market data have a lot of missing data gaps, especially for some of the emerging and developing economies.

A broader measure of financial development that includes the stock market could yield different results. Further studies are recommended for sector-specific studies as the positive relationship between FDI and EPU could have been driven by inflows to industries with strategic advantages. Financial technology might offer new ways to manage risk; thus, future studies are also recommended on how financial innovations (such as fintech, blockchain, and alternative financing) affect the interaction between EPU and FDI. Policymakers should seek to strengthen financial systems as they have a crucial role to play in moderating the effects of policy uncertainty and foreign direct investment.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Table A1. Panel unit root tests.

Variable	Cross-Sectional Dependence	None	Intercept and Trend	Intercept	Decision
Panel unit root test using the LLC					
EPU	21.51 ***	−7.89757 ***	−8.30240 ***	−9.42018 ***	I(0)
FD	71.59 ***	−27.63638 ***	8.77191 ***	−9.96308 ***	I(1)
FDI	23.08 ***	−7.09999 ***	−3.12065 ***	−5.30639 ***	I(0)
FD_EPU	51.41 ***	−37.54508 ***	48.72081 ***	−9.0825 ***	I(0)
Panel unit root test using the IPS					
EPU	21.51 ***	-	−6.14735 ***	−8.98614 ***	I(0)
FD	71.59 ***	-	−16.59173 ***	−18.53282 ***	I(1)
FDI	23.08 ***	-	−5.63928 ***	−8.08428 ***	I(0)
FD_EPU	51.41 ***	-	−21.91266 ***	−26.36007 ***	I(0)
Panel unit root test using the CIPS					
EPU	21.51 ***	-	−3.230 ***	−3.159 ***	I(0)
FD	71.59 ***	-	−4.127 ***	−4.048 ***	I(1)
FDI	23.08 ***		−3.490 ***	−3.045 ***	I(0)
FD_EPU	51.41 ***	-	−3.090 ***	−2.844 ***	I(0)

Source: authors' own calculations using EViews 14. *** $p < 0.001$. Source: authors' own computations using Stata 15.1.

Table A2. Sample of countries.

Country				
Albania	**Colombia**	**Indonesia**	**Moldova**	**Sierra Leone**
Angola	Costa Rica	Iraq	Mongolia	South Africa
Argentina	Cote d'Ivoire	Jamaica	Morocco	Sri Lanka
Armenia	Dominican Republic	Jordan	Mozambique	Sudan
Bangladesh	Ecuador	Kazakhstan	Myanmar	Tajikistan
Belarus	Egypt, Arab Rep.	Kenya	Namibia	Tanzania
Benin	El Salvador	Kyrgyz Republic	Nicaragua	Thailand
Bolivia	Ethiopia	Lao PDR	Niger	Tunisia
Bosnia and Herzegovina	Georgia	Lesotho	Nigeria	Tunisia
Brazil	Ghana	Madagascar	Pakistan	Turkiye
Burkina Faso	Guatemala	Malawi	Paraguay	Uganda
Burundi	Guinea	Malaysia	Peru	Ukraine
Cambodia	Guinea-Bissau	Mali	Philippines	Venezuela, RB
Cameroon	Honduras	Mauritania	Rwanda	Viet Nam
China	India	Mexico	Senegal	Zimbabwe

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